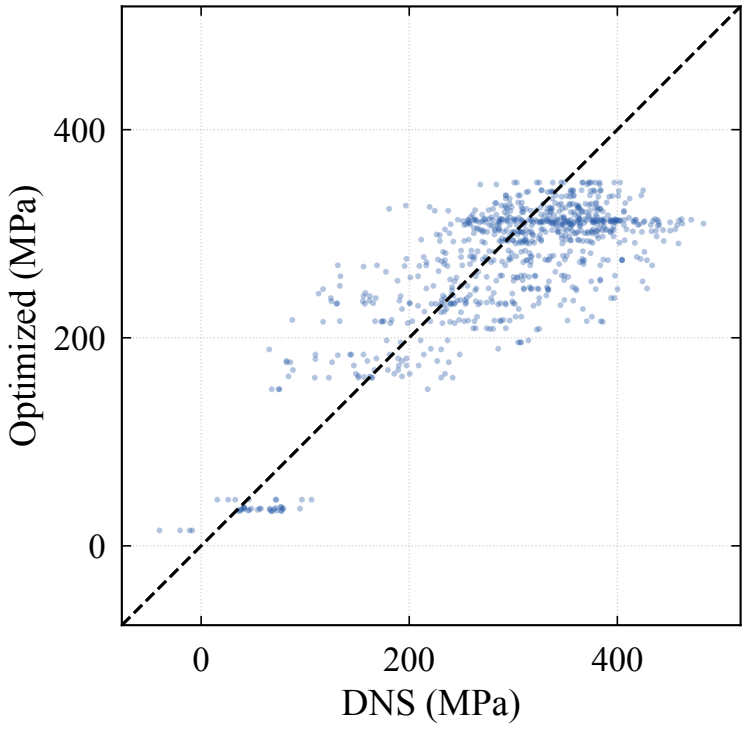
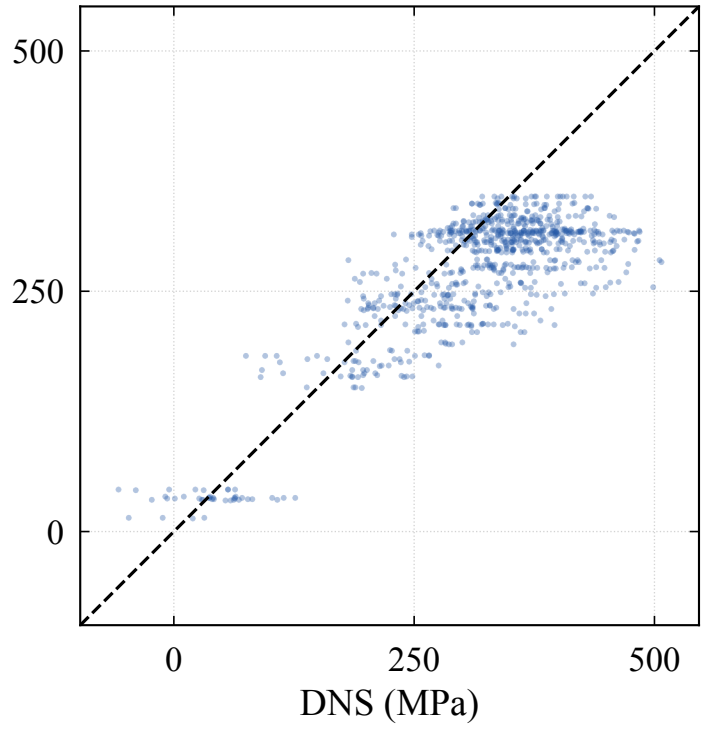
DNS vs optimized: paired-field agreement

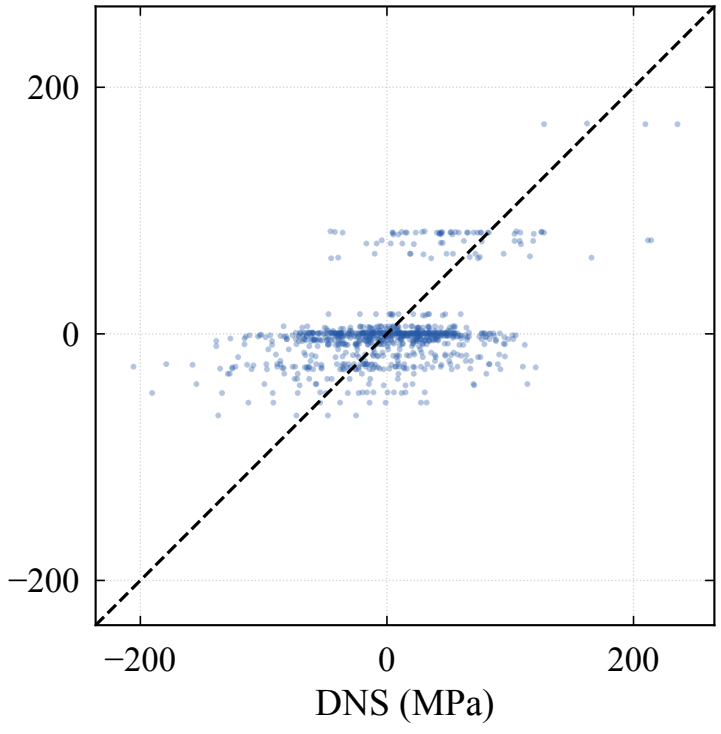
eL2 = 20.5% | R² = 0.55
RMSE = 63.2 MPa



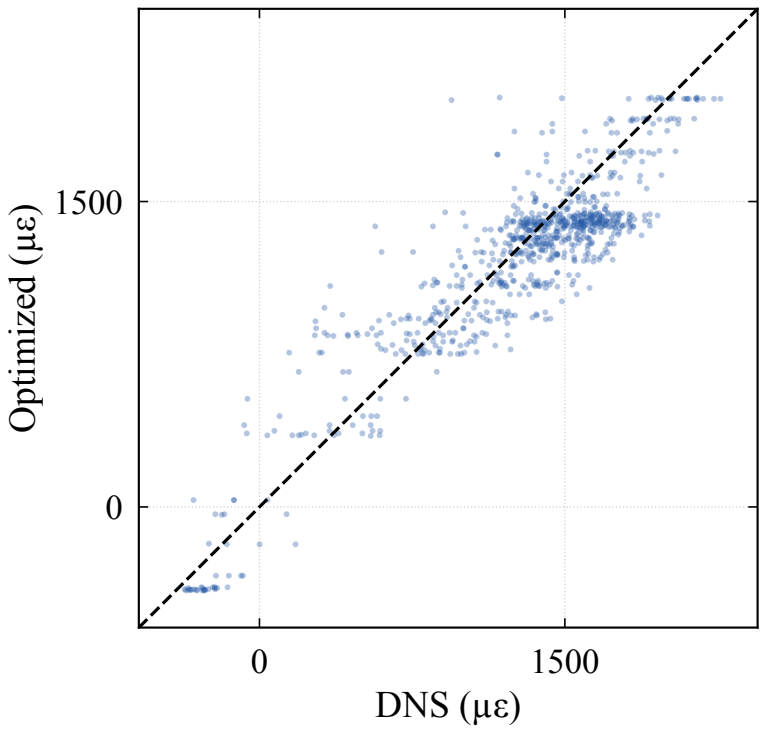
eL2 = 22.5% | R² = 0.40
RMSE = 74.4 MPa



eL2 = 90.8% | R² = 0.18
RMSE = 49.3 MPa

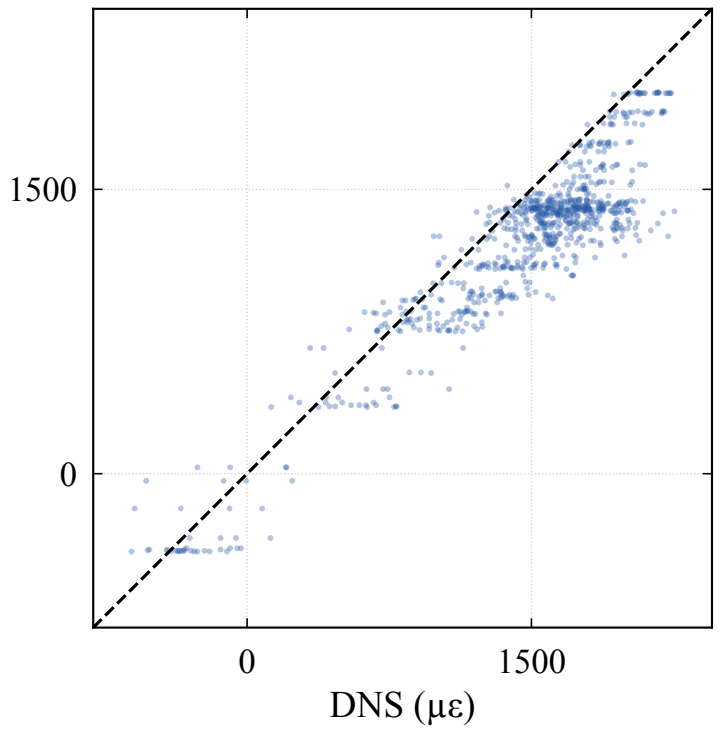


(a) Stress xx
eL2 = 16.9% | R² = 0.80
RMSE = 233 με



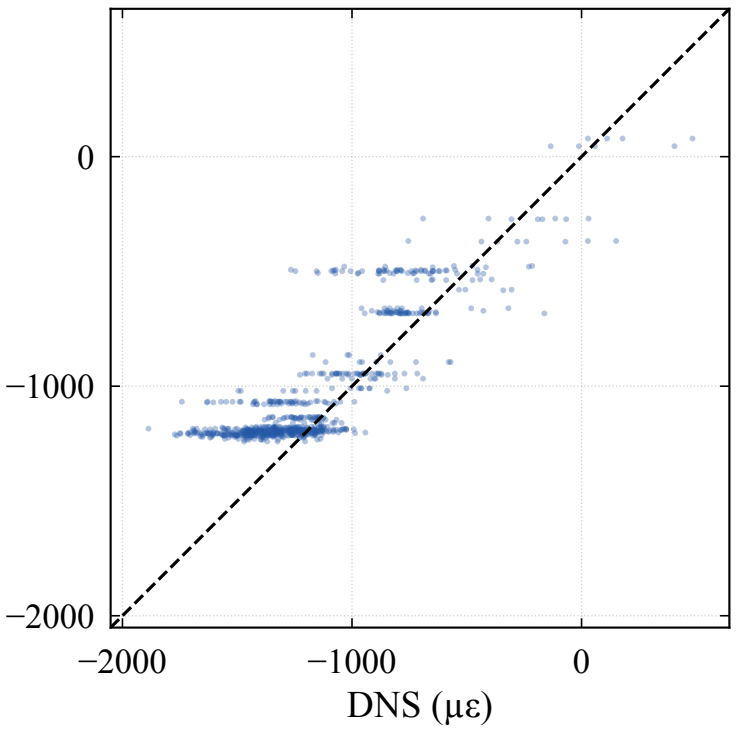
(d) Elastic strain xx

(b) Stress yy
eL2 = 21.1% | R² = 0.63
RMSE = 329 με

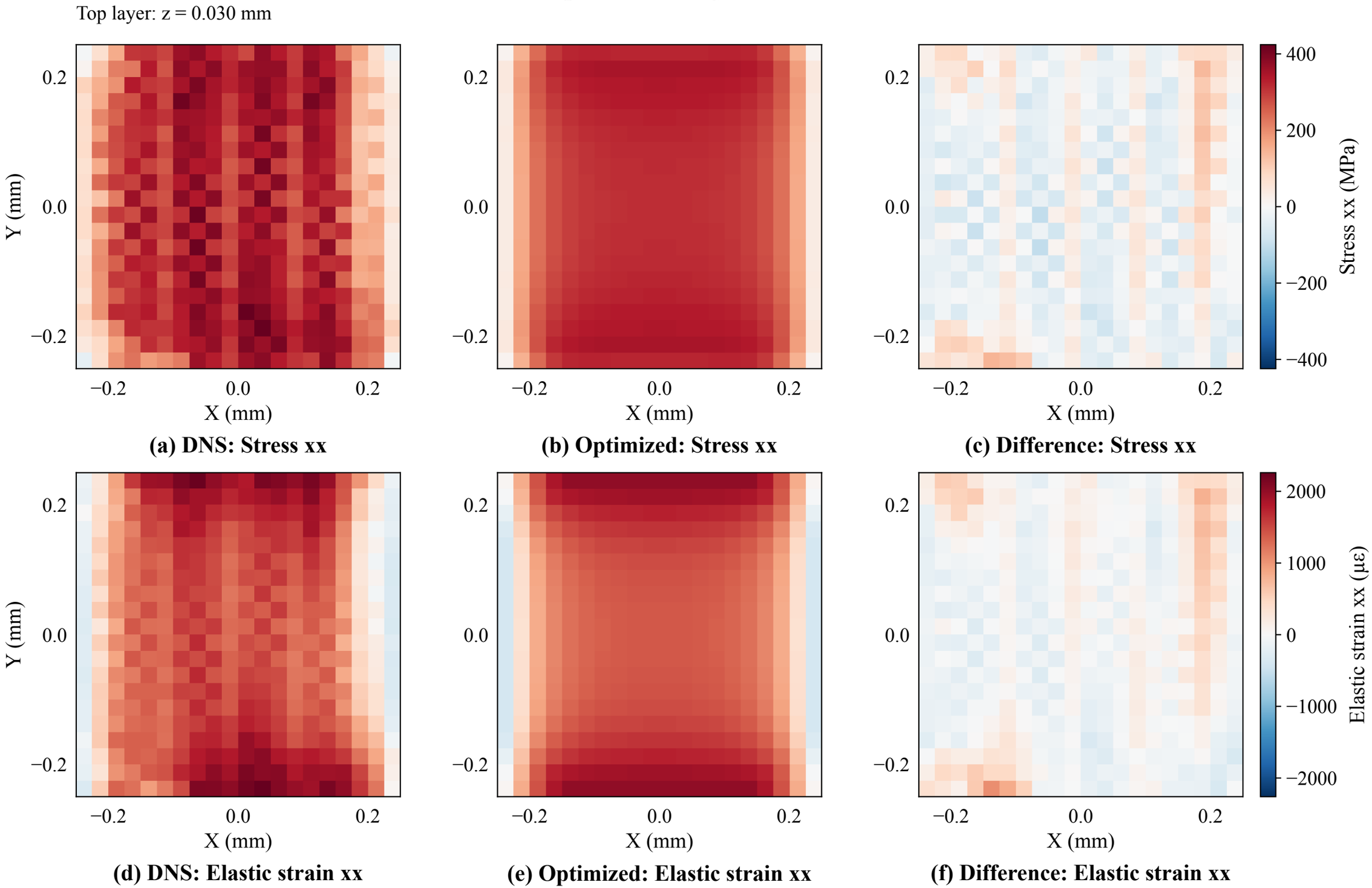


(e) Elastic strain yy

(c) Stress zz
eL2 = 18.4% | R² = 0.56
RMSE = 224 με

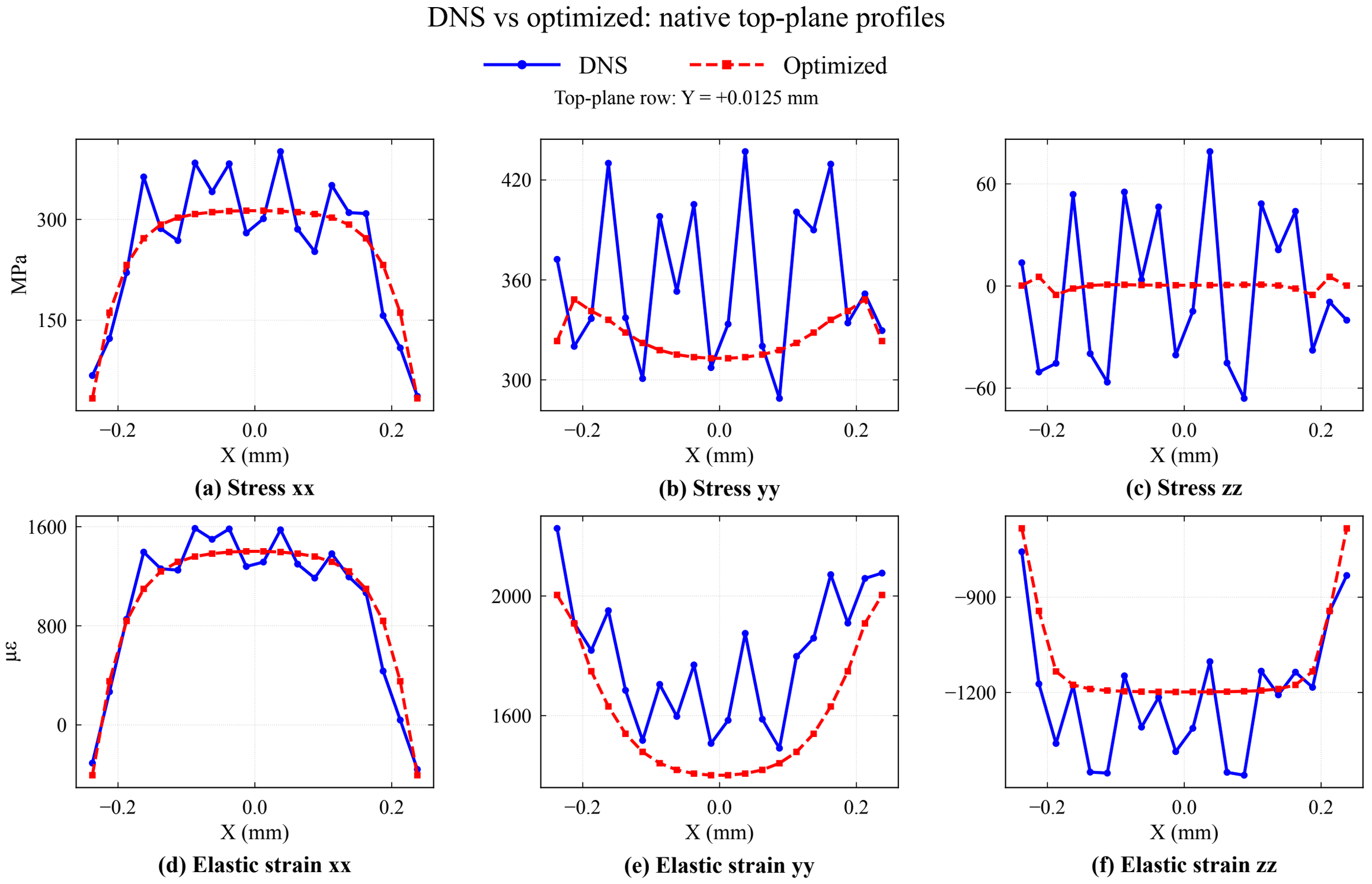


(f) Elastic strain zz



(a) DNS: Stress xx **(b) Optimized: Stress xx** **(c) Difference: Stress xx**

(d) DNS: Elastic strain xx **(e) Optimized: Elastic strain xx** **(f) Difference: Elastic strain xx**


DNS vs optimized: native top-plane profiles
DNS
Optimized
Top-plane row: Y = +0.0125 mm
MPa
300
150
420
360
300
60
0
−60
1600
800
0
2000
1600
−900
−1200
−0.2
0.0
0.2
X (mm)
(a) Stress xx
(b) Stress yy
(c) Stress zz
με
(d) Elastic strain xx
(e) Elastic strain yy
(f) Elastic strain zz